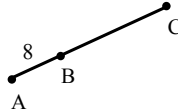


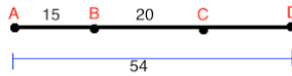
Students will use the Pythagorean Theorem to determine the distance between two points.

Do Now

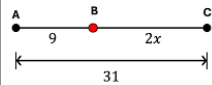
- 1) Solve for BC if $AC = 22$.



- 2) Solve for CD .

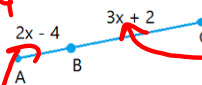


- 3) Solve for x .



- 4) Find BC if $AC = 8$. $BC = 3(2) + 2$
 $BC = 8$

$2(2) - 4$
 0



$AB + BC = AC$

$2x - 4 + 3x + 2 = 8$

$5x - 2 = 8$
 $\downarrow 2 \quad + 2$

$5x = 10$
 $\div 5 \quad \div 5$

$x = 2$

Do Next

- 1) Round the following numbers to the nearest nearest hundredth:

7.1091 _____

3.635 3.6

2.073 2.07

6.941 _____

12.107 _____

2.3845 _____

- 2) Round the following numbers to the nearest nearest whole number:

7.1091 _____

56.83 _____

2.073 _____

6.941 _____

12.42 _____

2.501 _____

- 3) Round the following numbers to the nearest nearest tenth:

19.825 _____

9.654 _____

0.222 _____

15.771 _____

16.230 _____

0.99 _____

- 4) Round the following numbers to the nearest nearest cent:

\$125.689 _____

\$1,340.196 _____

\$1.993 _____

- 5) Round the following numbers to the nearest nearest dollar:

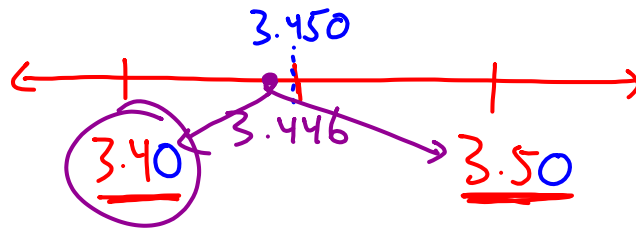
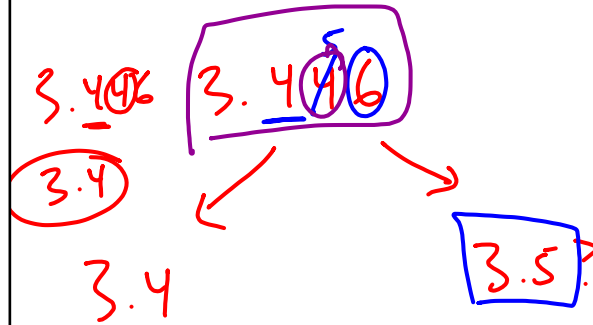
\$125.689 _____

\$1,340.196 _____

\$1.993 \$2.0

.

Round to nearest tenth.



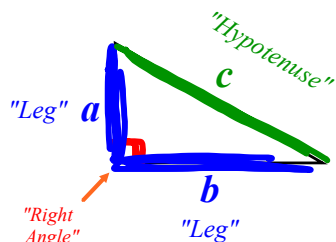
 https://www.google.com/search?sca_esv=68354a3c7c70afc3&udm=7&fbs=AlljpHxU7S

Pythagorean's Theorem

Any triangle that has a 90° angle is called a right Δ.

In a diagram, we represent the 90° right angle with a box.

Every right triangle is made up of 2 legs and 1 hypotenuse



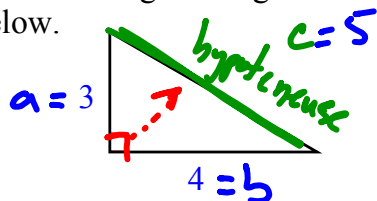
The Legs (a & b) are the two sides that touch the Right Angle (It doesn't matter which leg is a and which is b).

The Hypotenuse (c) is the only side that does not touch the Right Angle.

The Hyp. must be the longest side !

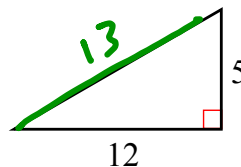
If we know the length of any two sides of a Right Triangle, Pythagorean's Theorem ($a^2 + b^2 = c^2$) allows us to find the length of the third side. To use this formula correctly, a & b must be the lengths of the Legs and c must be the length of the Hypotenuse!

Ex 1: Find the length of the hypotenuse in the right triangle shown below.



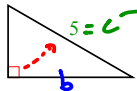
$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 (3)^2 + (4)^2 &= c^2 \\
 9 + 16 &= c^2 \\
 25 &= c^2 \\
 \sqrt{25} &= \sqrt{c^2} \\
 5 &= c \\
 \boxed{5} &= c
 \end{aligned}$$

Ex 2: Find the length of the missing side in the right triangle shown below.



$$\begin{aligned}
 (5)^2 + (12)^2 &= c^2 \\
 25 + 144 &= c^2 \\
 \sqrt{169} &= \sqrt{c^2} \\
 13 &= c
 \end{aligned}$$

Ex 3: Find the length of the missing leg in the right triangle below.



$$a^2 + b^2 = c^2$$

$$(3)^2 + (b)^2 = (5)^2$$

$$9 + b^2 = 25$$

$$-9 \quad -9$$

$$\sqrt{b^2} = \sqrt{16}$$

$$b = 4$$

Ex 4: Find the length of the missing side in the right triangle below.



$$a^2 + b^2 = c^2$$

$$a^2 + 8^2 = 17^2$$

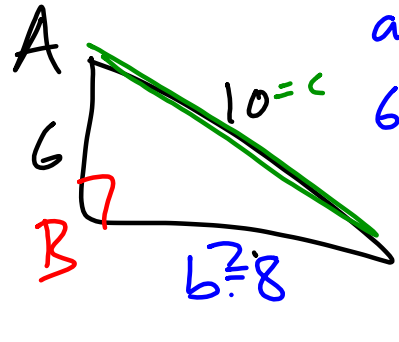
$$a^2 + 64 = 289$$

$$-64 \quad -64$$

$$\sqrt{a^2} = \sqrt{225}$$

$$a = 15$$

Ex 5: In right triangle ABC , angle B is a right angle. If $AB = 6$ and $AC = 10$, what is the length of BC ?



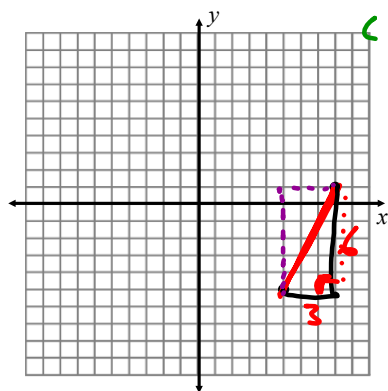
$$a^2 + b^2 = c^2$$

$$6^2 + b^2 = 10^2$$

$$b = 8$$

*geometry pro-tip: if a problem describes a diagram to you, draw the diagram

Ex 6: What is the distance between the points (8,1) and (5,-5), rounded to the nearest tenth?



$$a^2 + b^2 = c^2$$

$$3^2 + 6^2 = c^2$$

$$9 + 36 = c^2$$

$$\sqrt{45} = \sqrt{c^2}$$

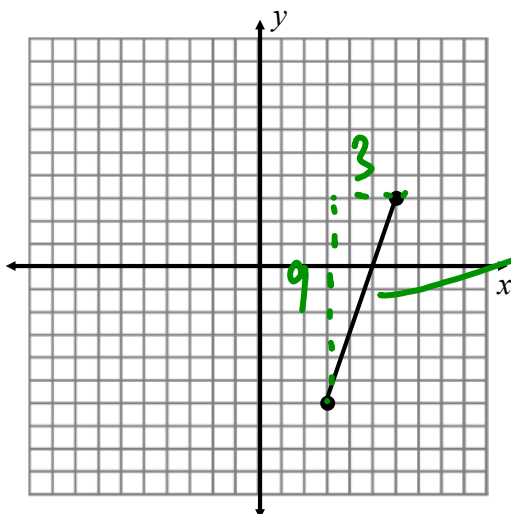
$$c = \sqrt{45}$$

$$c = 6.708...$$

$$\boxed{c \approx 6.7}$$

units

Ex 7: Find the length of the line segment shown in the graph below.
Round your answer to the nearest tenth.



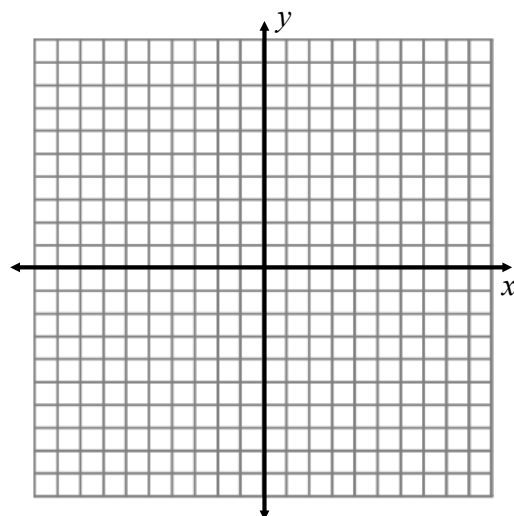
$$a^2 + b^2 = c^2$$

$$3^2 + 6^2 = c^2$$

$$9 + 36 = c^2$$

$$\sqrt{45} \approx \boxed{6.7}$$

Ex 8: The coordinate of the vertices of quadrilateral $ABCD$ are $A(1,2)$, $B(-5,3)$, $C(-6,-3)$ and $D(0,-4)$. Justin claims that $ABCD$ is a rhombus. Is Justin correct? Justify your response.



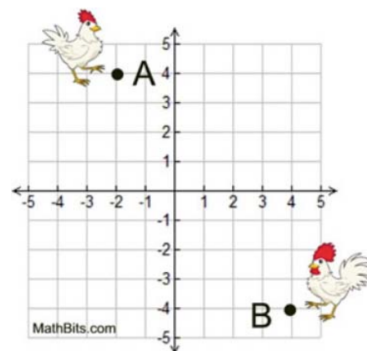
Key fact: A rhombus is a four-sided shape having all sides that are the same length.

Challenge Problems

Ex 9: A chicken and a rooster, on the same straight line, are heading toward a chicken coop that is halfway between them. The chicken is at point $A(-2,4)$ and the rooster is at point $B(4,-4)$.

a. What are the coordinates of the chicken coop?

b. How far will the chicken and the rooster each have to travel to reach the coop?



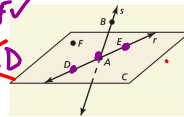
Ex 10: Given $A(-1,10)$ and $B(x,2)$. Find a value(s) of x such that the distance between A and B is exactly 10.

Assignment 1.3: Measuring Segments

Use the diagram to the right to answer 1 - 3.

1. Give two other names for line DE and plane C .

Plane DEF ✓
~~Plane AED~~

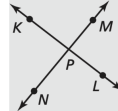


2. Name three points that are collinear. Name four points that are coplanar.

3. Name a point that is *not* coplanar with points D , B , and E .

Use the diagram to the right to answer 4 - 5.

4. Give another name for KL .

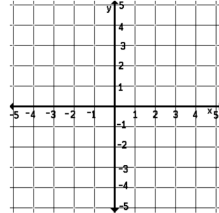


5. Are ray KP and ray PK the same ray? Are ray NP and ray NM the same ray? Explain your reasoning.

6. Explain why a four-legged chair may rock from side to side even if the floor is level. Would a three-legged chair on the same level floor rock from side to side? Why or why not?

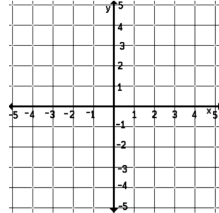
Plot the given points, then determine if the following segments are congruent. Show work.

7. $A(-2, 4)$, $B(3, 4)$, $C(0, 2)$, $D(0, -2)$



$\overline{AB}$ and $\overline{CD}$ are _____

8. $A(-5, 5)$, $B(-5, -2)$, $C(-4, 3)$, $D(3, 3)$



$\overline{AB}$ and $\overline{CD}$ are _____

For 9 - 10, find FH .

9. F 8 G 14 H

10. F 37 H 13 G

For 11 - 12, find RS .

11. R $2x+10$ S $x-4$ T
21

12. R $4x-9$ S 19 T
 $8x-14$

